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26 February 1982
SUPERSEDING
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5 August 1968

MILITARY SPECIFICATION

SEMITRAILER CHASSIS: INTERMODAL CARGO CONTAINER CARRYING, COMMERCIAL

AND

CONTAINER, CARGO: INTERMODAL, DEMOUNTABLE, COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 20 and 40 foot semitrailer chassis and 20 and 40 foot demountable closed cargo containers (intermodal type) for transporting and storage cargo. The semitrailer chassis and demountable containers may be procured as a combined unit or as individual items to supplement an existing system, as specified in 1.2. The semitrailer chassis are of the tandem axle type and have minimum gross weight ratings of 51,050 pounds for the model A, 20 foot chassis, 74,500 pounds for the model B, 40 foot chassis, and 75,800 pounds for the model C, 40 foot chassis. Items procured under this specification are commercial items which shall be warranted by the manufacturer as specified in procurement documents.

1.2 Classification. The following models of equipment shall be provided, as specified (see 6.2):

Model A - Semitrailer chassis, 20 foot cargo container carrying
Model B - Semitrailer chassis, 40 foot cargo container carrying

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GBS.l, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposa1 (DD Form 1426) appearing at the end of this document or by letter.

FSC-2300

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Model C - Semitrailer chassis, one 40 foot or two 20 foot cargo container carrying
Model D - Container, cargo, demountable, 20 foot
Model E - Container, cargo, demountable, 40 foot.

2. APPLICABL DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handboods. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS
MILITARY

MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1223	- Non-Tactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking, and Data Plate Standards.
MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28-Volt, Waterproof.

2.12 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington D.C. 20591.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI Designation MH5.1.1M-1979	- Requirements for Closed Van Cargo Containers.
ANSI Designation MH5.6-1978	- Basic Interface Requirements for Cargo Container Chassis.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SAE Inc.

SAE Standards and Recommended Practices.

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Truck, Truck-Tractors, and Trailers.
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J682	- Rear Wheel Splash and Stone Throw Protection.
J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers.
J702	- Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer.
J1292	- Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring.

(Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.

Yearbook (latest edition).

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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(Industry association specif1cations and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard semitrailer, container, and accessories. Except as specified in 3.1.1 through 3.1.1.4, the semitrailer chassis, containers, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All semitrailer chassis and container items shall be as represented in the semitrailer chassis and container manufacturer's technical data, and special or mounted equipment shal1 be as represented in the equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of semitrailer chassis and container models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items.

3.1.1 Special requirements. In addition to the standard semitrailer chassis, container, and components specified in 3.1, the semitrailer chassis and container shall be furnished with special provisions and equipment as specified herein.

3.1.1.1 Painting and marking. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking, and data plates shall be in accordance with MIL-STD-1223.

3.1.1.1.1 Semitrailer chassis marking. Each semitrailer chassis shall be marked indicating the maximum payload rating (gross weight minus curb weight) expressed in pounds and kilograms. Ratings shall be indicated by stenciling, or on a decal or permanently attached plate, and shall be visible with container(s) locked in travel position on the chassis.

3.1.1.1.2 Container marking. Each container shall be marked indicating tare weight expressed in pounds and kilograms, loaded weight capacity expressed in pounds and kilograms, and shipping cube expressed in cubic feet and cubic meters. The data shall be indicated by stenciling, or on a decal or permanently attached plate, and shall be applied to each side of the container.

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3.1.1.1.3 Special markings. When specified by the procuring activity (see 6.2), special markings shall be applied to the item(s).

3.1.1.2 Rustproofing. When specified (see 6.2), model A, B and C semitrailer chassis shall be rustproofed in accordance with MIL-STD-1223.

3.1.1.3 Wood treatment. When specified (see 6.2), wood shall be treated in accordance with the requirements of MIL-STD-1223.

3.1.1.4 Lifting and tying down attachments. The semitrailer chassis shall be provided with lifting and tying down attachments conforming to MIL-STD-209, type III. Attachments shall be designed for lifting and tying down the chassis in an empty configuration.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards In effect on the date of manufacture.

3.2 General design.

3.2.2 Interchangeability and compatibility. Dimensions and relative positions of all assembled mating parts of units produced by the contractor shall conform to this specification and shall insure the following:

(a) Interchangeability of all semitrailer chassis and containers
(b) Functional compatibility and proper coupling between each semitrailer chassis and container(s)

3.2.3 Dimensions, weights and ratings. Dimensions, weights and ratings of the semitrailer chassis and containers shall conform to 3.2.3.1 and 3.2.3.2 and shall be as specified herein. Component and vehicular ratings shall not be raised to meet the requirements of this specification.

3.2.3.1 Semitrailer chassis. Model A, B, and C semitrailer chassis shall have the dimensions, weights and ratings specified in table I.

TABLE I. Semitrailer chassis dimensions, weights and ratings.

Characteristic	Model A	Model B	Model C
Overall length (maximum), feet	24	41 1/2	41 1/2
Overall width (maximum), inches	96	96	96
Curb weight (maximum), pounds	6,250	7,229	8,581
Gross weight rating (minimum), pounds	51,050	74,500	75,800

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3.2.3.2 Containers. Model D and E containers shall have the dimensions, weights and ratings specified in table II.

TABLE II. Container dimensions, weights and ratings.

Characteristic	Model D	Model E
Overall length, nominal, feet	20	40
Overall width (maximum), inches	96	96
Overall height (maximum), inches	102	102
Tare weight (maximum), pounds	5,270	9,660
Gross loaded weight rating (minimum), pounds	44,800	67,200

3.3 Performance.

3.3.1 Semitrailer unit. The complete semitrailer unit (chassis and container(s)), fully equipped and loaded to its gross weight rating (see 3.2.3.1), shall be capable of being towed at speeds as great as 25 miles per hour (mph) over unimproved roads and over reasonably hard uneven terrain; and being towed at speeds as great as 55 mph over improved roads.

3.3.1.1 Turning ability. The semitrailer unit shall be capable of assuming a 90 degree turn angle to the coupled towing vehicle without cramping or damaging the semitrailer or the towing vehicle

3.3.1.2 Tracking ability. The semitrailer chassis, unloaded and loaded as specified, shall conform to the tracking requirements of Federal Motor Carrier Safety Regulation 393.70(a).

3.3.1.3 Brake performance. The service brakes shall stop the semitrailer-tractor combination, with the semitrailer unit loaded to its gross weight rating, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.3.2 Containers. The cargo containers shall meet all dimensional and strength design requirements of ANSI MH5.1.1M-1979. Certification of conformance to ANSI requirements shall be furnished (see 4.3.3 and 4.3.3.1).

3.4 Semitrailer chassis components.

3.4.1 Frame. The semitrailer chassis frame shall be the manufacturer's standard all steel frame with longitudinal members supporting the running gear, landing gear and upper fifth wheel plate. The flanges of the main frame members on straight frame chassis shall be one piece

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without splices or joints. Where frame members are spliced on gooseneck chassis, joints shall be reinforced to fulfill the dynamic loading requirements of 3.3.1. Splices shall be designed to avoid stress concentration. Splice welds shall be continuous. The frame shall support the weight of the fully loaded container(s) specified in table II. Model A, B and C semitrailer chassis shall have the dimensions and clearances specified in table III. Container-chassis interface shall be in accordance with ANSI MH5.6-1978.

TABLE III. Semitrailer chassis dimensions and clearances.

Description	Model A	Model B	Model C
Ground clearance (minimum), inches	14	14	14
Upper fifth wheel height, from ground to underside of upper fifth wheel plate, inches, plus or minus one inch	48	48	48
Distance from vertical centerline of kingpin to front end of semitrailer (minimum), inches	18	36	36
Swing radius from vertical centerline of kingpin to most difficult point of semitrailer nose (maximum), inches	60	60	60

3.4.1.1 Bolsters. The semitrailer chassis frame shall include full-width, steel front and rear bolsters. Model C semitrailer chassis shall, in addition, include a mid-center, full-width, steel bolster. Rear bolsters may serve as rear crossmembers. Twist lock and pin lock devices specified in 3.4.1.2 shall be installed at the bolster ends. Bolsters shall be reinforced as necessary in accordance with manufacturer's standard commercial practice.

3.4.1.2 Locking devices. The semitrailer chassis shall be equipped with twist-lock devices on the frame bolsters to engage and secure the lower corner fittings of the container(s). Model A and B chassis shall be provided with four twist-lock devices, one at each corner of the frame front and rear bolsters, or with two pin-lock devices on the front bolster of the frame and with two twist-lock devices on the rear bolster of the frame. Model C chassis shall be provided with four twist-lock devices, one at each corner of the frame front and rear bolsters and with four retractable twist-lock devices at the mid-center bolster for securing two 20 foot containers. The locking devices shall be designed to preclude unintentional release. The locking devices and their locations shall be in accordance with ANSI MH5.6-1978, and the devices shall meet the requirements of Federal Motor Carrier Safety Regulation 393.100(e).

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3.4.2 Suspension system. The semitrailer chassis shall be equipped with a suspension system, with components having a rated capacity at least equal to the load imposed upon each member, measured at the ground, with the semitrailer unit loaded to its gross weight rating. When suspension is rated at the spring pads, unsprung weight shall be deducted.

3.4.3 Axles. The semitrailer chassis sha11 be provided with tandem axles. Through shaft axles shall be furnished. Axle spindles shall be of steel, hardened and ground to size. Wheel bearings shall be of the antifriction roller bearing type. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground with the semitrailer loaded to its gross weight rating. The wheel bearings and axle spindles shall be oil lubricated. The hubs shall be fitted with seals. The hub caps shall have a window and shall incorporate provisions for venting or an equivalent method of withstanding pressures without leakage.

3.4.4 Wheels, rims, tires and tubes. The semitrailer chassis shall be equipped with dual wheels on each axle. Rims and tire ratings shall conform to the Tire and Rim Association recommendations for the type and size tires furnished. Rim sizes shall be the same for all wheels on the semitrailer. Tire size and load range (ply rating) shall be the same for all tires on the semitrailer chassis. When specified (see 6.2), disc type wheels shall be furnished.

3.4.4.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of the steel belted radial ply type or, when specified (see 6.2), bias ply. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the semitrailer unit loaded to its gross weight rating.

3.4.4.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided in accordance with Tire and Rim Association recommendations.

3.4.4.3 Carrier for spare tire assembly. Unless otherwise specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the curbside, forward of the axles. The carrier shall permit removal or installation of a spare wheel assembly, using only tools furnished with the semitrailer chassis.

3.4.4.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be mounted and secured on the carrier.

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3.4.4.5 Spare tire. When specified (see 6.2), an inflated spare tire shall be mounted on the spare wheel or rim. The spare tire shall be of the same size, load range, and tread design as the tires in use on the semitrailer chassis.

3.4.5 Brakes. Full air, internal expanding type brakes shall be furnished. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43, 393.45 through 393.47, and 393.52. The brake system shall include piping, hose connections, gladhands, spring loaded dust covers or dummy couplings equipped with security chains, provisions for stowing dummy couplings, relay emergency valve, air reservoirs with a remote control drain located above the bottom of the frame rails, slack adjusters, brake chambers, and all other parts and connections required for a complete air brake system. Gladhands shall conform to SAE J318. The location of the gladhands shall be as recommended in SAE J702. All components of the brake system shall be installed in a manner to provide road clearance for travel over uneven or rough terrain and protection from objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of the wheel rims.

3.4.5.1 Parking brakes. The semitrailer chassis shall be equipped with spring or air diaphragm mechanical lock type parking brakes. The parking brakes shall hold the semitrailer unit, loaded to its gross weight rating, on a 10 percent grade despite the depletion of the compressed air supply. The parking brakes shall be automatically applied upon disconnection off the emergency air line and under emergency conditions. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

3.4.6 Upper fifth wheel plate. The upper fifth wheel plate shall be designed for coupling to full oscillating and fore and aft rocking type fifth wheels, shall be of sufficient size to cover a fifth wheel 36 inches in diameter, and shall conform to Federal Motor Carrier Safety Regulation 393.70(b). A 2-inch kingpin fabricated of heat-treated alloy steel and conforming to SAE J700 shall be furnished. The forward end of the upper fifth wheel plate shall have a turned-up lip for ease of coupling and for body protection.

3.4.7 Landing gear. The semitrailer chassis shall have two vertical lift, telescopic, nonrotating, landing legs with two-speed gears, and a hand crank on the curbside. Landing legs shall be provided with steel wheels or self-leveling skid pads. A holder shall be provided for the hand crank when not in use. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to the portion of the gross weight sustained, and the additional forces resulting from impact during coupling and uncoupling operations. When

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placed in the travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to preclude the entrance of foreign matter which would impair its functioning or mechanical efficiency. The semitrailer unit, loaded to its gross weight rating (see 3.2.3.1), shall not nose over when resting on the landing gear.

3.4.7.1 Range of adjustment. The landing gear shall have a range of adjustment sufficient to vary the height of the upper fifth wheel plate at the kingpin from 46 inches to not less than 54 inches from the ground. The minimum ground clearance with the landing gear retracted shall be 12 inches at a 48-inch upper fifth wheel plate height.

3.4.8 Rear end protection. Semitrailer unit rear end protection shall be provided in accordance with Federal Motor Carrier Safety Regulation 393.86.

3.4.9 Lighting. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of rub rails (unless recessed and fully protected) or on the semitrailer bumper. Turn signal lamps conforming to SAE J588 shall be furnished at the rear of the semitrailer. All electrical wiring shall conform to SAE J1292. Clearance and identification lights shall be constructed for easy removal and replacement of lamps and lenses without the use of hand tools.

3.4.9.1 Receptacle 12-volt. The front of the semitrailer chassis shall be equipped with a 7-contact receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein. The receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

3.4.9.2 Twelve-volt direct current (dc) system. A 12-volt dc lighting system conforming to 3.4.9 and 3.4.9.1 shall be provided. The 12-volt dc lighting system shall conform to Federal Motor Carrier Safety Regulations 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32, and 393.33.

3.4.9.3 Independent 24-volt dc system. When specified (see 6.2), a 24-volt dc lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.4.9. The

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24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.4.9.2, except the 24-volt dc stop and turn lamps shall be combined.

3.4.9.4 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the 12-volt dc system (see 3.4.9.2) from a towing vehicle equipped with a nominal 24-volt dc electrical system shall also be provided.

3.4.9.5 Receptacle, 24-volt dc. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer chassis shall also be equipped with a l2-contact receptacle and cover conforming to MS75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B.	Connect to left-hand turn signal and stop lamp (yellow).
Contact D.	Connect to ground (white).
Contact E.	Connect to clearance, side marker, identification, and tail lamps (black and brown).
Contact J.	Connect to right-hand turn signal and stop lamp (green).
Contact L.	Connect to ground (white).

The rest of the contacts shall not be connected. Circuits B and J on tactical trucks are combination stop and turn indicator circuits. On the interconnected 24-volt system, the normal l2-volt dc turn signal lights will function both as turn signals and stop lights, and the normal l2-volt dc stop lights will not be operational when the semitrailer chassis is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt dc, l2-contact receptacle.

3.4.9.6 Resistors. When an interconnected 24-volt dc system is specified (see 3.4.9.4), the 12-contact receptacle shall be provided with resistance in each circuit to reduce the voltage of the tactical (military design) towing vehicle from a nominal 24 volts dc to 12 volts dc. Each circuit resistor shall be selected to reduce the 28-volt dc regulated voltage of the tactical truck to be compatible with the maximum rated voltage of the semitrailer chassis electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a beat sink to preclude overheating and any damage to resistors, wiring, or adjacent components.

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3.4.11 Mud flaps. Manufacturer's standard mud flaps shall be provided to the rear of the rear wheels in accordance with SAE J682.

3.5.1 Container components.

3.5.1 Construction and materials. The design, construction, and materials shall be as specified herein. Material shall be free from all defects and imperfections that might adversely affect the serviceability and appearance of the finished product. The design and construction of the containers shall conform to the requirements for closed cargo containers specified in ANSI MH5.1.1M-1979. Dimensions, tare weights and ratings shall conform to table II and table IV.

TABLE IV. Container dimensions.

Description	Model D	Model E
Interior dimensions, (minimum):		
Length, inches	231	472 1/2
Width, inches	92	92
Height, inches	92 1/2	92 1/2
Door opening height, (minimum) inches	90	90
Door opening width, (minimum) inches	90	90
Capacity, (minimum) cubic feet	1,137	2,327

3.5.1.1 Container frame. The container frame shall be the manufacturer's standard steel or combination steel and aluminum construction. Structural members shall be reinforced at joints to fulfill the dynamic and static loading requirements specified herein. Steel structural members shall be arc welded. Aluminum structural members shall be arc welded or riveted. Effective means shall be taken to preclude electrolytic action between dissimilar metals.

3.5.1.2 Sidewall configuration. The sidewalls and front wall shall be of either exposed post, smooth skin, or corrugated skin construction. When smooth skin construction is used, the sidewall and front wall skins shall be smooth and flat (without corrugations, but exposed panel lap splices are allowable for metal skin). When exposed post construction is used, sidewall skin sheets shall form a smooth continuous surface inside the container (laps at skin panel splices are permitted). When corrugated skin construction is used, skin panels shall be joined and butt welded. Post recesses, except at front and rear corner posts, shall be covered. The interior of the sidewalls and front wall for all types of construction, except fiberglass, fiberglass reinforced plywood, or encapsulated plywood construction, shall be lined with 114 inch thick exterior grade fir plywood.

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3.5.1.3 Floor construction. The container floor shall either be minimum 1 1/8 inch thick laminated hardwood boards, composite type floor constructed of minimum 1 7/64 inch thick hardwood boards and steel flanges (hat shaped channels) extending the full length of the container, or minimum 1 7/64 inch thick plywood may be used. The longitudinal board edges shall be adjoined by tongue-and-groove or ship-lap joints. Attachment to the crossmembers shall be by means of at least two fasteners per board per crossmember. Countersunk head fasteners, 5/16 inch diameter, either of the self-tapping screw type or machine screws with self-locking nuts shall be installed so that each head is 0 to 1/16 inch below the board surface and at least one inch from the board edge. For composite type floor construction, floorboards shall be accurately fitted between the hat shaped channels and between side members and hat shaped channels. The boards shall be secured to the bottom flanges of the hat shaped channels with l/4-inch minimum diameter, self-tapping corrosion resistant screws, two at each cross sill and two on each side of each butt joint. Screw heads shall be countersunk. The edges of the boards shall be sealed against entry of water from the underside of the floor. The floor shall be installed to permit lateral variations in floor board width due to swelling while also prohibiting the entrance of water from the underside. The flooring shall not be installed tight against the outer longitudinal rail. The flooring shall be capable of withstanding concentrated loads imposed by forklifts having a 14,000 pound axle load supported by 7-inch wide tires spaced 30 inches apart on the tire centerline.

3.5.1.4 Roof construction. The roof assembly shall be constructed to insure drainage. The edge attachment shall be so designed to prevent, water intrusion into the container through the fasteners, and to allow free drainage of water from the roof surface. All roof seams shall be weatherproof.

3.5.1.5 Doors. Two doors shall provide a clear width opening for access into the container. Doors shall be hung within the rear end structure with not less than three heavy duty pin hinges on each door, permitting the doors to fold back approximately 270 degrees against the sides of the container. Each door shall be provided with handle operated cam locking devices. Means shall be provided to hold doors in the full open position. Door gaskets shall be installed to insure a seal that will prevent water entry.

3.5.1.5.1 Anti-pilferage provisions. Door hinge and locking devices shall be attached with fasteners offering minimum access for removal with common tools from the outside surface of the door. Additionally, exposed conventional hexagonal nuts on the inner door surface shall not be used. Where installation of "tee" nuts on the inside of the door is not practical, the bolt heads

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may be inside, secured outside by conventional hex nuts, and nuts tightened and tack welded to the bolts. Each hinge pin shall be permanently secured to the hinge butt by cold heading or tack welding the pin in place. Where welding destroys protective plating or hardware items, the weld and surrounding area shall be thoroughly cleaned and primed with one coat of zinc chromate and at least one coat of finish paint with a dry film thickness of not less than 1.5 mils. All locking device handles shall be furnished with provisions for padlocking.

3.5.1.6 Container corner fittings. The container corner fittings shall conform to ANSI MH5.1.1M-1979.

3.5.1.6.1 Container corner fitting securement devices. When specified (see 6.2), four securement devices shall be furnished for securing together containers which are stacked for storage or transport. Securement devices shall be either stacking and locking cone type or stacking and pin locking type.

3.5.2 Document holders. Weatherproof documentation pockets or boxes shall be provided on both the inside and outside of the right rear door of each container. The inside box or pocket shall be designed to hold a package of electric adding machine (EAM) cards, approximately 3/4 inch thick, and the service or operator's manual. The outer box or pocket shall provide visual access to the Government Bills of Lading (GBL's) and other shipping documents without their removal. The approximate size of each box or pocket shall be 9 by 12 inches.

3.6 Stress analysis. The contractor shall furnish a stress analysis of his proposed semitrailer chassis construction. This analysis shall include at least the calculations and diagrams required herein.

(a) The sprung load distribution diagram with the vehicle loaded to its gross weight rating.
(b) The vertical shear diagram for the length of the chassis.
(c) The bending moment diagram for the length of the chassis
(d) The loads, sections, and stresses in members that carry maximum shear and moment, to include the analysis of the stresses in attaching weldments or other fastenings.
(e) The loads, sections, shear, and bending moment diagram at kingpin and rear underconstruction subframe members.
(f) The load, section, shear, and bending moment diagram for the most critical crossmember, to include the analysis of stresses in attaching weldments or other fastenings.
(g) The loads, sections, and stresses in members that carry maximum compression.

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(h) A listing of all assumptions and references utilized to provided required data.

3.7 Servicing and adjusting. Prior to acceptance by the Government inspector, the contractor shall service and adjust each semitrailer chassis for operational use including at least the following: adjustment of brake system; inflation of all tires; and complete lubrication with grades of lubricants recommended for the ambient air temperature at the delivery point. The semitrailer chassis shall be conspicuously tagged to identify the lubricants and their temperature range.

3.8 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

3.8.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and clipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing and sawing. Re-entrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.8.2 Fiberglass. Fiberglass or fiberglass coated plywood used in the fabrication of equipment shall be smooth and free from bubbles, glazing cracks, and discontinuities. All edges of fiberglass reinforced plywood shall be completely sealed to prevent water intrusion between the laminates.

3.8.3 Bolted and riveted connections. Bolt and rivet holes shall be accurately punched or drilled and shall have the burrs removed. Washers, lockwashers, or lock nuts shall be provided; and all bolts, nuts, and screws shall be tight. Rivet heads, when not countersunk or flattened, shall be of uniform size and shape for the same diameter of rivet, concentric with the rivet holes, and in full contact with the surface of the members.

3.8.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or

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other foreign matter. Weld penetration shall be such as to provide adequate transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration.

3.8.5 Machined work. All parts shall be manufactured to gage through the use of proper jigs and fixtures and shall provide interchangeability of parts as manufactured.

3.8.6 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or defects which might render the castings unsound for use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government verification. Quality assurance operations performed by the contractor shall be subject to Government verification at unscheduled intervals. Verification shall consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first production semitrailer chassis and container procedure under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine conformance to the requirements of the contract. Acceptance of the first production semitrailer chassis and container shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Vehicle weight. The first production semitrailer chassis shall be weighed to determine curb weight and distribution of curb weight on kingpin and axles. The imposed loading on the kingpin and axles will be computed using curb weight, container weight, and

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payload uniformly distributed. The first production container shall be weighed to determine tare weight.

4.3.2 Road test. The first production semitrailer chassis shall be road tested for a distance of not less than 50 miles, with and without container(s) loaded to the semitrailer gross weight rating, to determine conformance to the requirements of this specification.

4.3.3 Container certification. Manufacturer's certification that the container furnished meets the requirements of ANSI MH5.1.1M-1979 shall be furnished. The first production container shall be tested for weatherproofness as specified in ANSI MH5.1.1M-1979.

4.3.3.1 Availability of standards. The contractor shall make available to the Government representative, at the first production vehicle inspection, a current copy of ANSI MH5.1.1M-l979 and ANSI MH5.6-1978. The documents shall be surrendered to the Government for the personal use of the Government representatives in conjunction with the contract for the duration of the contract.

4.3.4 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.5 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action bas been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein.

4.5 Wood treatment certification. Manufacturer's records, acceptable to the Government, shall be furnished to verify that all wood requiring treatment in accordance with MIL-STD-1223 has been treated.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

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6. NOTES

6.1 Intended use. The semitrailer chassis and containers covered by this specification are intended for nontactical use by the Government for transporting cargo. The containers are not intended for transporting ammunition. Additionally, the containers covered by this specification are intended for use by the Government for transporting cargo by sea (not air or rail).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Model and nomenclature of equipment (see 1.2).
(c) Appropriate service for treatment, painting, marking and data plates (see 3.1.1.1).
(d) Special markings, if required (see 3.1.1.1.3).
(e) Rustproofing, if required (see 3.1.1.2).
(f) Wood treatment, if required (see 3.1.1.3).
(g) Disc type wheels, if required (see 3.4.4).
(h) Bias ply tires, if required (see 3.4.4.1).
(i) If a spare tire carrier is not required (see 3.4.4.3).
(j) Spare wheel or rim, if required (see 3.4.4.4).
(k) Spare tire, if required (see 3.4.4.5).
(l) Independent 24-volt system, if required (see 3.4.9.3).
(m) Interconnected 24-volt system, if required (see 3.4.9.4).
(n) Container securement devices, if required (see 3.5.1.6.1).

6.3 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.4 Changes in classification. This specification formerly covered a 20 foot coupleable semitrailer chassis, and a 20 foot demountable container body capable of being coupled into a 40 foot unit. This revision deletes the coupled configuration and offers, instead, three models of semitrailer chassis for 20 foot, 40 foot and two 20 foot containers and two containers, a 20 foot and a 40 foot size.

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